Cost-Effective Hybrid Sensor Network Optimization for Flood Monitoring in Resource-Constrained Environments Using Genetic Algorithms

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Abstract *- Flood early-warning systems in resource-constrained environments face persistent limitations related to sparse hydro-meteorological observations, unreliable communications, limited energy availability, and severe financial constraints. Although recent advances in low-cost Internet of Things (IoT) sensing and wireless sensor networks (WSNs) demonstrate the feasibility of decentralized flood monitoring, existing studies typically optimize either sensor placement, network routing, energy efficiency, or prediction accuracy in isolation. Consequently, current approaches rarely provide an integrated framework capable of simultaneously addressing coverage, reliability, redundancy, resilience, and lifecycle cost. This paper presents a systematic review and methodological framework for cost-effective hybrid sensor network optimization for flood monitoring using Genetic Algorithms (GAs) and Non-dominated Sorting Genetic Algorithm II (NSGA-II). The article synthesizes findings from AI-enabled flood modelling, multi-modal river monitoring, self-healing WSN architectures, and budget-constrained optimization. Building upon these insights, the paper proposes a formalized multi-objective optimization framework in which each chromosome simultaneously encodes a hybrid sensor portfolio and geographic deployment coordinates. Unlike conventional weighted-sum approaches, the proposed framework reformulates the optimization problem as a constrained Pareto-based search using NSGA-II to preserve trade-offs between flood coverage, reliability, redundancy, and cost. To address scale incompatibility between heterogeneous objectives, a dynamically normalized Sensor Validity Function (SVF), inspired by the Weighted Sum Validity Function (WSVF), is incorporated as an auxiliary scalar benchmark. The framework further integrates constrained domination, diversity-preserving niching mechanisms, and local greedy refinement strategies to improve convergence and deployment diversity. The proposed framework contributes a publication-oriented methodological foundation for resilient and economically feasible flood-monitoring infrastructures in developing regions. To provide computational validation of the proposed NSGA-II framework, this paper presents a real-world case study applied to the Kur River in Sabirabad District, Azerbaijan, using a heterogeneous 15-sensor fleet subject to a $6,000 budget constraint. Experimental results demonstrate that NSGA-II returns a Pareto front of 100 non-dominated solutions compared to a single solution produced by the weighted-sum GA benchmark, achieving 100% monitoring coverage versus 96.7% for GA, reducing deployment cost by $800, and yielding a hypervolume indicator 1.94× larger than the single-objective baseline.*

Keywords: *Flood Monitoring, NSGA-II, Multi-objective Optimization, Wireless Sensor Networks (WSN), Cost-Effective Deployment, Genetic Algorithms.*

I. Introduction

Floods remain among the most destructive natural hazards globally, causing severe economic losses, infrastructure damage, and human displacement, particularly in developing countries where monitoring infrastructure and emergency response capabilities are limited. The challenge of designing effective flood-monitoring systems in such environments extends beyond hydrological sensing alone. Practical deployments must simultaneously account for sparse communication infrastructure, unstable power availability, difficult terrain accessibility, budget limitations, and dynamic environmental disruptions during disaster events.

Recent developments in wireless sensor networks (WSNs), low-cost IoT platforms, and artificial intelligence (AI)-based flood prediction systems demonstrate the potential for scalable and decentralized monitoring architectures. However, most existing research addresses only isolated dimensions of the problem. Some studies focus primarily on energy efficiency, others emphasize routing reliability or prediction accuracy, while a separate body of work investigates deployment cost reduction. Consequently, relatively few studies provide a unified optimization framework that jointly considers sensor selection, geographic placement, resilience, redundancy, and operational cost.

Evolutionary optimization techniques, particularly Genetic Algorithms (GAs), have emerged as effective tools for solving complex combinatorial optimization problems involving conflicting objectives. Their suitability arises from their ability to search large non-convex spaces without requiring gradient information while simultaneously supporting multi-objective optimization. Nevertheless, many flood-monitoring optimization studies continue to rely on scalar weighted-sum formulations that collapse inherently conflicting objectives into a single fitness value. Such formulations often obscure trade-offs between cost and monitoring performance and introduce significant sensitivity to manually selected weighting parameters.

This paper addresses these limitations by reframing flood sensor deployment as a constrained multi-objective optimization problem solved using NSGA-II [7]. The proposed framework integrates concepts from elitist evolutionary optimization, normalized validity aggregation, and niching-based diversity preservation to support resilient and economically feasible sensor deployments.

The contributions of this paper are fourfold:

A systematic synthesis of literature spanning AI-based flood modelling, WSN optimization, self-healing disaster networks, and low-cost sensor systems.

A formalized chromosome representation that jointly encodes sensor portfolio composition and spatial placement.

A mathematically rigorous NSGA-II optimization framework with normalized objective functions, constrained domination, and diversity-preserving operators.

A hybrid methodological architecture integrating Pareto optimization, dynamic normalization, niching mechanisms, and local placement refinement for flood-monitoring deployments in resource-constrained environments.

The paper therefore functions not only as a review article but also as a methodological framework paper proposing a rigorous optimization architecture ready for adoption in simulation and real-world deployment studies.

A. AI and Machine Learning for Flood Risk Modelling

Jones et al. (2023) demonstrate that AI-based flood modelling can significantly accelerate flood prediction and hazard assessment through data-driven surrogate modelling, uncer-tainty quantification (UQ), and geospatial analytics. Their study shows that machine learning (ML) models trained using approximately 10,000 data points can perform flood prediction considerably faster than computationally expensive hydro-dynamic simulations. However, the authors also report systematic overprediction caused by Digital Elevation Model (DEM) limitations and insufficient representation of flood-defense infrastructure such as sea walls.

These findings are highly relevant for sensor-network optimization because they highlight the importance of strategically placed in-situ sensors serving as ground-truth calibration anchors. In practical flood-monitoring systems, sensors are not merely passive measurement devices; they also function as calibration points capable of reducing structural modelling errors in AI-based flood prediction systems.

The study further emphasizes the importance of balancing computational speed and predictive accuracy rather than optimizing one objective in isolation. This observation directly motivates the use of evolutionary optimization frameworks capable of simultaneously balancing multiple competing objectives. The use of surrogate modelling and uncertainty-aware parameter exploration also supports the integration of global optimization techniques such as GAs and NSGA-II for identifying robust deployment configurations under uncertainty.

From a methodological perspective, the implications are significant. Flood-monitoring optimization should not only maximize hydrological coverage but also prioritize sensor locations that improve model calibration quality, uncertainty reduction, and resilience to incomplete environmental information.

B. Multi-Modal River Monitoring in Poorly Gauged Basins

The feasibility of real-time multi-modal monitoring in data-scarce regions is further validated by Mdegela et al. (2023), who implemented automated systems measuring rainfall, river levels, and atmospheric conditions in the Kikuletwa River basin. Their findings emphasize that effective flood monitoring in poorly gauged environments requires moving beyond purely hydrological metrics to incorporate operational realities. Specifically, the study highlights that optimal sensor placement is a multi-dimensional decision influenced by infrastructure accessibility, communication stability, and maintenance feasibility.

A critical insight from this practical deployment is that a sensor network optimized solely for hydrological coverage may result in configurations that are economically unsustainable or physically inaccessible during peak flood events. For instance, the strategic placement of sensors near existing infrastructure, such as bridges, significantly enhances installation stability and operational oversight. These real-world constraints-ranging from connectivity instability to lifecycle maintenance costs-provide a direct empirical justification for a multi-objective optimization framework.

Consequently, the design of a resilient monitoring network necessitates a balanced trade-off between competing dimensions: monitoring coverage, communication reliability, redundancy, and lifecycle cost. This convergence of practical and technical requirements naturally motivates the transition toward Pareto-based evolutionary optimization, such as NSGA-II, which can explicitly navigate these conflicting deployment priorities.

C. Low-Cost IoT Flood Monitoring and Deployment Challenges

Recent advancements in affordable IoT platforms, particularly those utilizing NodeMCU, ESP8266 and HC-SR04 ultrasonic sensors, have shifted the paradigm of flood early-warning systems. As demonstrated by Nik Mohd Kamal et al. (2025), these decentralized solutions offer a viable, high-frequency alternative to prohibitively expensive traditional hydrological stations. While their findings confirm that cloud-integrated cloud-based data streaming is technically feasible at

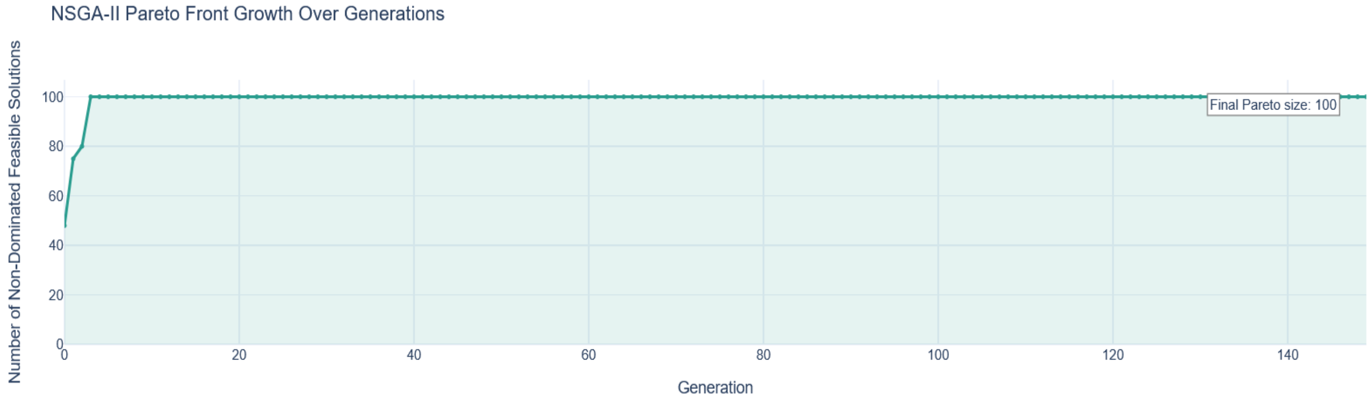


Figure 1. Theoretical trade-off between flood monitoring coverage and system cost. The red curve defines the Pareto-optimal frontier, representing solutions where coverage cannot be increased without a corresponding increase in budget. Point A indicates a cost-effective solution, while the "Infeasible Region" represents performance targets that are unattainable under current resource constraints.

a fraction of the usual cost, a critical operational challenge remains unaddressed.

The transition from "hardware feasibility" to "field-ready deployment" lacks a rigorous methodology. Specifically, existing literature often overlooks the complex trade-offs between sensor modality selection, precise geographic positioning, and the strategic allocation of network redundancy. In resource-constrained environments, simply reducing hardware costs is insufficient if the deployment strategy fails to account for signal connectivity and long-term maintenance accessibility.

This paper bridges the gap between low-cost sensing and optimized network architecture. By consolidating sensor selection and spatial coordinates into a unified evolutionary chromosome representation, the proposed framework ensures that affordability does not come at the expense of hydrological reliability or network resilience.

D. Evolutionary and Bio-Inspired Networking Approaches

Energy-Efficient Hybrid Heuristics

Ullah et al. (2024) investigate energy-efficient WSN architectures by integrating clustering strategies with Fuzzy Spider Monkey Optimization and Hidden Markov Models (HMMs). Their research demonstrates that hybrid heuristics significantly extend network longevity through optimized cluster-head selection. However, a persistent limitation in these methods is the underlying assumption of pre-existing sensor nodes. The fundamental challenge of determining the optimal count, modality, and spatial distribution of sensors remains largely unaddressed.

Self-Healing Networks in Disaster Scenarios

The resilience of disaster-oriented sensor networks is critical, as highlighted by Ahmed et al. (2024). Their hybrid BOA-PSO architecture enables self-healing capabilities, allowing routing structures to adapt dynamically to catastrophic disruptions. This paradigm is particularly relevant for flood-monitoring environments where sensor displacement or failure is frequent. Consequently, a robust optimization framework must account for both initial deployment and potential event-time reconfiguration to maintain network integrity.

Evolutionary Clustering and Routing

Priyadharshini et al. (2022) emphasize the balance between residual energy and communication efficiency through biogeography-based optimization. While their approach improves inter-cluster organization, it shares a common gap with other routing-centric studies: it treats deployment as a static prerequisite. There is a clear need for a unified model that jointly optimizes the sensor portfolio and geographic coordinates to ensure operational efficiency from the outset.Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced. Styles named “Heading 1”, “Heading 2”, “Heading 3”, and “Heading 4” are prescribed.

E. RESEARCH GAP AND METHODOLOGICAL MOTIVATION

The existing literature on flood-monitoring optimization exhibits three fundamental methodological limitations that necessitate a more robust computational approach.

Objective Fragmentation

Current studies frequently isolate specific network components-such as routing efficiency or energy harvesting-while overlooking the broader operational context. By ignoring the coupling between geographic deployment feasibility and total lifecycle costs, these fragmented models fail to provide actionable infrastructure designs for resource-constrained environments.

Limitations of Weighted-Sum Formulations

A significant portion of prior research simplifies multi-objective problems into a single scalar fitness function $F(X)$, typically expressed as:

$$F(x) = w_1 \cdot Coverage + w_2 \cdot Reliability + w_3 \cdot Redundancy - w_4 \cdot Cost$$

Although computationally straightforward, this weighted-sum approach introduces three critical vulnerabilities:

Firstly, dimensional incompatibility arises because objectives such as coverage, measured on a fractional scale between 0 and 1, and cost, measured in monetary units, exist on vastly different numerical scales. Without rigorous normalization, the objective with the larger numerical magnitude tends to dominate the search space disproportionately.

Secondly, weight sensitivity occurs because the optimization outcome depends heavily on manually assigned weights w_i, which are frequently arbitrary and lack a clear objective justification.

Thirdly, the loss of Pareto information occurs because scalarization reduces the multi-dimensional trade-off surface to a single solution point, thereby limiting the ability of decision-makers to explore the Pareto-optimal frontier where conflicting objectives, such as maximizing coverage and minimizing cost, can be effectively balanced. The fundamental difference between a single-objective weighted-sum approach and the proposed multi-objective optimization is illustrated in Figure 1.

Formalization and Reproducibility

There is a notable lack of formalization regarding chromosome encoding and constraint-handling mechanisms in existing flood-monitoring literature. This ambiguity hinders the reproducibility of results and the scalability of models to real-world river basins.

II. Methods

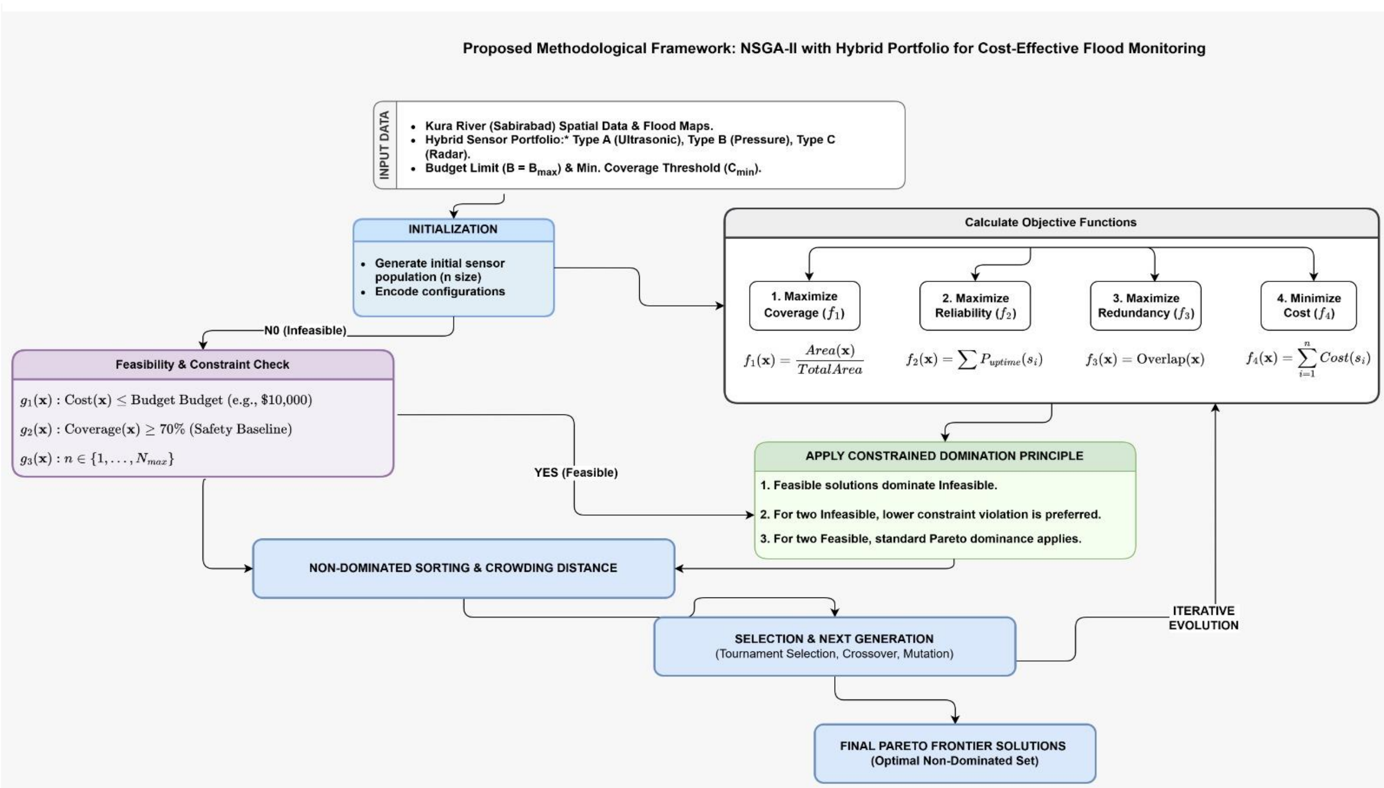
A. Proposed Methodological Framework

To overcome these gaps, this paper introduces an integrated optimization architecture based on the Non-dominated Sorting Genetic Algorithm II (NSGA-II). The proposed framework diverges from traditional methods by implementing:

Pareto-based multi-objective search enables the exploration of the solution space without collapsing multiple objectives into a single optimization function, thereby preserving important trade-offs between conflicting goals. Dynamic normalization is achieved through a specialized Sensor Validity Function (SVF), which ensures scale neutrality across heterogeneous metrics and prevents dominance by objectives with larger numerical ranges.

Diversity preservation is maintained through niching and crowding-distance mechanisms that sustain a broad range of alternative deployment configurations and reduce premature convergence. Local Greedy Refinement is enhancing global evolutionary search with localized geographic placement logic for improved spatial accuracy. The complete optimization pipeline of the proposed framework is illustrated in Figure 2.

Figure 2. Proposed methodological framework for NSGA-II-based hybrid sensor network optimization for cost-effective flood monitoring. The pipeline proceeds from spatial and sensor portfolio inputs through population initialization, multi-objective function evaluation (coverage f_1, reliability f_2, redundancy f_3, and cost f_4), feasibility and constraint checking, and constrained-domination sorting, before iterating through non-dominated ranking, crowding-distance computation, and tournament selection with crossover and mutation operators. Infeasible solutions are recycled through the constraint check, while feasible solutions advance to the constrained-domination stage. The process repeats across successive generations until convergence, yielding a final Pareto-optimal non-dominated set that represents the full spectrum of cost–performance trade-offs available to decision-makers.



B. *Proposed Methodological Framework*

Problem Formulation

Let a candidate deployment solution be represented as a set of parameters:

$$x = \{(s_i, x_i, y_i,)\}_{i=1}^{n}$$

where s_1 denotes the specific sensor type (selected from a hybrid portfolio), (x_i, y_i) represents the geographic coordinates within the study area (the Kur River segment), and n is the total number of sensors deployed.
The primary objective of this framework is to identify an optimal deployment configuration that maximizes flood-monitoring effectiveness while simultaneously minimizing total economic and operational expenditures. Unlike conventional scalar optimization models, this framework treats the deployment task as a constrained Multi-Objective Optimization Problem (MOOP), recognizing that coverage, reliability, and cost are often conflicting dimensions.

Multi-Objective NSGA-II Formulation

The optimization problem is formally defined by the following objective functions, which seek to balance performance with fiscal responsibility:

Maximize Coverage is area monitored relative to the total flood-prone zone.

$$f_1\,(x) = Coverage(x)$$

Maximize Reliability is weighted probability of continuous system uptime.

$$f_2(x) = Reliability(x)$$

Maximize Redundancy is overlap in sensing to ensure resilience against node failure.

$$f_3(x) = Redundancy(x)$$

Minimize Cost is total capital and operational expenditure.

$$f_4(x) = Cost(x)$$

To align with the standard NSGA-II minimization architecture, the problem is formulated as:

$$Minimize\ F(x) = \{-f_1(x), -f_2(x), -f_3(x), -f_4(x)\}$$

Subject to the following constraints:
The optimization process is subject to several operational constraints. The budgetary constraint, $g_1(x)$, requires that the total deployment cost does not exceed the maximum allocated budget B. The performance threshold constraint, $g_2(x)$, ensures that the achieved coverage remains above the minimum acceptable level C_{min}, thereby maintaining a baseline standard of monitoring safety and effectiveness. In addition, the deployment limit constraint, $g_3(x)$, restricts the number of deployed sensors to the range $n \in \{0,1,2, \dots N_{max}\}$, reflecting the physical and practical limitations of the network scale.
By maintaining these objectives separately, the framework preserves the Pareto Frontier, providing decision-makers with a range of optimal trade-offs rather than a single, potentially biased scalar solution.

C. *Constrained Domination Principle*

To handle the constraints effectively without the need for sensitive penalty parameters, the framework implements the Constrained-Domination Principle. Within this mechanism, a solution i is said to *constrained-dominate* solution j if any of the following conditions are met:

A solution is considered superior under three primary conditions. First, solution i is preferred if it is feasible while solution j is infeasible, giving feasibility the highest priority in the optimization process. Second, if both solutions are infeasible, preference is given to the solution with the smaller overall magnitude of constraint violation, thereby encouraging progression toward feasibility. Third, when both solutions are feasible, Pareto superiority determines preference, meaning that solution i dominates solution j by performing no worse in any objective and strictly better in at least one objective within the objective space.

Figure 3. Spatial mapping of the optimized chromosome. The plot shows the actual coordinates (x_i,y_i) and specific types (s_i∈{A,B}) for 10 deployed stations along the Kur River, demonstrating the integration of variable-length chromosome encoding and spatial diversification.

This approach ensures that the search process prioritizes feasible regions of the search space while naturally converging toward the Pareto-optimal set under strict budget and coverage limitations.

D. Chromosome Representation and Genetic Operators

Chromosome Encoding

To effectively address the optimization problem, we adopt a hybrid chromosome structure that simultaneously encodes sensor selection and spatial distribution. The chromosome C is represented as:

$$C = [s_1, s_2, \dots, s_n \mid x_1, y_1, x_2, y_2, \dots , x_n, y_n]$$

Within this framework, $s \in \{A, B, C\}$ represents the specific type assigned to each sensor, while (x_i, y_i) denotes the spatial deployment coordinates of the sensor within the monitored environment. Furthermore, $n \in [n_{min}, n_{max}]$ enables variable chromosome length, providing flexibility in the total number of deployed sensors and allowing the optimization process to adapt the network scale according to system requirements and constraints.

By integrating the sensor portfolio and geographic placement into a single unified search structure, this formulation surpasses traditional deployment models that typically treat selection and positioning as decoupled tasks. This formulation extends traditional deployment optimization approaches by integrating both sensor selection and placement within a single search structure. The spatial result of this unified optimization is illustrated in Figure 3, which displays the specific coordinates and types for 10 optimized sensor stations along the Kur River corridor.The crossover mechanism is designed to respect the functional units of the model. To maintain the structural integrity of the individual sensors, crossover points are restricted to complete sensor triplets (s_i, x_i, y_i). This "unit-aware" approach ensures that sensor types remain linked to their respective coordinates, thereby preventing the generation of infeasible offspring that would otherwise result from splitting interdependent genes.

Mutation Operator

Global search diversity is maintained through the application of two distinct mutation operators:
Coordinate Mutation operator facilitates spatial exploration by perturbing existing coordinates using a Gaussian distribution:

$$(x_i', y_i') = (x_i, y_i) + \mathrm{N}(0, \sigma^2)$$

Here, σ is calibrated to 5% of the total basin extent, allowing for meaningful local adjustments without destabilizing the overall deployment pattern.

Sensor-type Mutation promotes portfolio diversification by reassigning a sensor's type according to a uniform distribution:

$$s_i' \sim Uniform\{A, B, C\}$$

Together, these operators balance the need for precise spatial refinement with the necessity of exploring new sensor combinations across the search space.

E. Formal Objective Functions

The optimization framework is driven by a multi-objective mathematical model designed to balance competing operational and economic requirements. To evaluate the fitness of each candidate solution, four distinct objective functions are formulated, representing coverage, reliability, redundancy, and lifecycle costs.

Spatial Coverage and Hydrological Priority

The primary goal of the sensor network is to maximize the monitored fraction of the river basin. In this model, coverage is defined not merely as a binary detection variable but as a weighted fraction of monitored river segments (R). The coverage objective is expressed as:

$$Coverage(x) = \frac{1}{|R|} \sum_{r \in R} \max_{i: s_i \ covers \ r} w(r)$$

By incorporating the term $w(r)$, which represents the hydrological importance weight of a specific segment, the formulation ensures that the genetic algorithm prioritizes critical zones-such as those with higher flood risks or proximity to human settlements-over less significant areas.

Network Reliability and Operational Feasibility

Beyond simple coverage, the long-term utility of the network depends on its operational resilience. The reliability function integrates two critical factors: signal connectivity and physical accessibility for maintenance. This relationship is quantified through a weighted average:

$$Reliability(x) = \frac{1}{n} \sum_{i=1}^{n} [\alpha \cdot Connectivity(x_i, y_i) + (1 - \alpha) \cdot Accessibility(x_i, y_1)]$$

System Redundancy and Fault Tolerance

To mitigate the risk of data loss due to individual sensor failure, a redundancy function is introduced to reward fault-tolerant configurations. Unlike the coverage function, which seeks to monitor each segment at least once, the redundancy objective incentivizes overlapping monitoring areas:

$$Redundancy(x) = \frac{1}{|R|} \sum_{r \in R} \frac{\min(2, |\{i: s_1 \ covers \ r\}|)}{2}$$

This mathematical structure specifically targets critical river segments, assigning higher fitness scores to solutions that provide double-coverage. By capping the reward at two sensors, the model prevents excessive resource clustering while ensuring a necessary safety margin for high-priority regions.

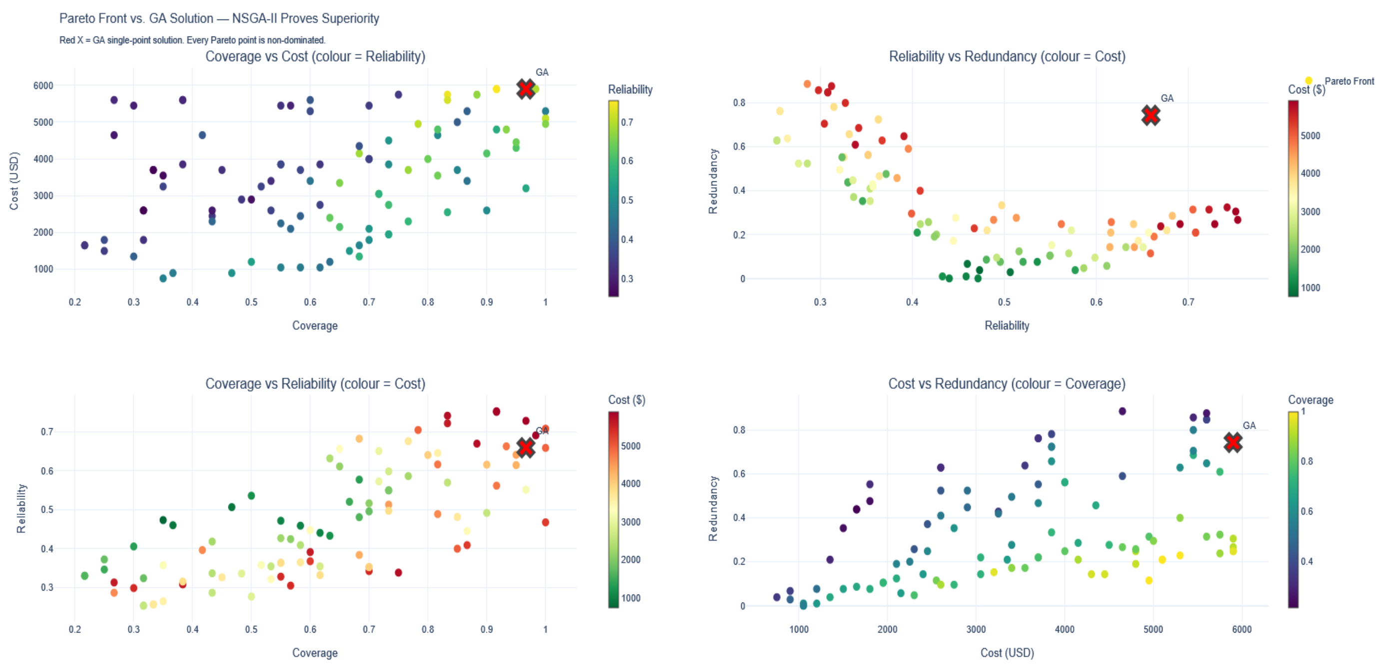


Figure 4. Multi-objective optimization results (Pareto Front). The scatter plot demonstrates the trade-off between coverage, reliability, and total cost, where each point represents an optimal network configuration that cannot be improved in one objective without degrading another.

Comprehensive Lifecycle Cost Analysis

Finally, the economic viability of the network is assessed through a total lifecycle cost function, following the budget-constrained optimization principles established in prior hybrid energy studies [9]. This objective accounts for the initial capital expenditure and the long-term operational costs associated with the chosen sensor portfolio over a planning horizon (T):

$$Cost(x) = \sum_{i=1}^{n} [ProcCost(s_i) + DeployCost(x_i, y_i) + MainCost(s_i) \cdot T]$$

This function integrates procurement expenses based on sensor type (s_i), installation costs dictated by the difficulty of the deployment coordinates (x_i, y_i), and recurring maintenance. By treating cost as a primary objective, the algorithm effectively navigates the Pareto front to identify high-performance configurations that remain within resource-constrained budgets.

F. Dynamic Normalization and Sensor Validity Function

To ensure numerical stability and parity across diverse performance metrics, the proposed framework implements a dynamic normalization strategy. A significant challenge in multi-objective optimization arises when objective functions operate on incompatible numerical scales; without intervention, direct aggregation tends to bias the evolutionary process toward high-magnitude objectives, such as total system cost, at the expense of lower-magnitude but equally critical parameters.

To mitigate this scaling bias, the methodology incorporates a dynamic normalization process inspired by the Weighted Sum Validity Function (WSVF) [8]. Each objective is transformed into a dimensionless unit through the following expression:

$$f_i'(x) = \frac{f_i(x) - f_i^{min}}{f_i^{max} - f_i^{min}}$$

In this formulation, f_i^{max} and f_i^{min} represent the extrema observed within the current population, which are updated dynamically throughout the evolution to reflect the shifting search space. These normalized values are subsequently integrated into a composite Sensor Validity Function (SVF), defined as:

$$SVF = \frac{1}{4}[f_1'(x) + f_2'(x) + f_3'(x) + (1 - f_4'(x))]$$

By aggregating these terms, the SVF provides a unified scalar baseline that facilitates comparative evaluation while functioning as a normalized indicator of deployment quality. This approach offers a more robust aggregation mechanism than conventional weighted-sum formulations, as it effectively neutralizes scale dominance. Consequently, the resulting SVF enhances the interpretability of the optimization results, ensuring that each performance metric contributes proportionally to the final fitness evaluation.

G. Niching and Diversity Preservation

To maintain the robust exploratory capabilities of the optimization process, the proposed framework incorporates specific niching and diversity preservation mechanisms. Traditional Genetic Algorithms (GAs) are frequently susceptible to premature convergence, often collapsing toward a single deployment strategy that may represent a local optimum. In the context of flood-monitoring networks, however, multiple near-optimal configurations often exist, each offering substantially different operational advantages. To prevent the loss of these diverse candidate solutions, the methodology integrates a niching strategy modeled after Hybrid Niching Genetic Algorithms (HNGA), ensuring that the population explores a broad spectrum of deployment architectures simultaneously.

A primary component of this diversity preservation is the implementation of the Selecting Factor Group (SFG), which governs mating restrictions. By facilitating crossover primarily among chromosomes characterized by similar sensor counts and portfolio structures, the framework minimizes the occurrence of "destructive crossover"-a phenomenon where the recombination of incompatible deployment architectures yields offspring with significantly degraded fitness. This structural alignment is further reinforced by the Comparing Factor Group (CFG) mechanism, which dictates that replacement competition occurs predominantly between structurally similar individuals. This intra-group competition ensures that unique deployment strategies are not prematurely eliminated by globally dominant but structurally unrelated solutions.

Complementing these global search mechanisms is a stage of Greedy Local Refinement, which functions as a fine-grained exploitation tool. Following the generation of offspring, individual sensor coordinates undergo a localized perturbation process. Each potential displacement is evaluated based on its immediate contribution to coverage improvement; beneficial adjustments are retained, while those that degrade performance are discarded. This localized optimization serves as an exploitation phase analogous to k-means refinement in hybrid clustering algorithms, allowing the framework to polish individual solutions and ensure that sensor placements are precisely tuned to the nuances of the local geographic terrain.

H. NSGA-II Optimization Procedure

The optimization workflow of the proposed framework follows a structured, evolutionary sequence designed to systematically explore and exploit the deployment decision space. The procedure begins with the initialization of a population of deployment chromosomes, where each individual represents a unique spatial configuration of sensors mapped across the target geographic coordinates. Following initialization, each candidate solution is rigorously evaluated against the predefined objective functions-maximizing coverage, reliability, and redundancy while minimizing implementation cost-alongside relevant operational constraints. To establish a multi-objective hierarchy, the population undergoes fast non-dominated sorting, partitioning solutions into distinct Pareto fronts based on their dominance status. Concurrently, a crowding distance metric is computed for each individual to quantify its density in the objective space, ensuring that subsequent selection mechanisms favor sparser, less explored regions of the front.

Parent selection is executed via crowded tournament selection, which balances fitness and diversity by prioritizing solutions with lower non-domination ranks or higher crowding distances when ranks are identical. Genetic variation is introduced through the application of crossover and mutation operators, generating an offspring population that explores new structural configurations. Immediately following this global exploration phase, the offspring undergo localized exploitation via the greedy placement refinement mechanism to fine-tune individual sensor positions. The parent and offspring populations are then merged into an expanded pool, maintaining elitism by ensuring that high-performing historical solutions are not lost. This combined pool is subjected to constrained domination sorting, allowing the algorithm to isolate and retain the elite non-dominated solutions while pruning sub-optimal configurations. This cycle repeats iteratively until convergence criteria are met, ultimately yielding a diverse Pareto front that represents the optimal trade-offs among the competing network objectives.

I. Evaluation Metrics

To assess the performance and reliability of the proposed framework, a multi-faceted evaluation suite is utilized, encompassing both algorithmic efficiency and structural robustness. The primary benchmark for optimization quality is the Hypervolume (HV) indicator, which quantifies the total volume of the objective space dominated by the obtained Pareto front relative to a predefined reference point. This is complemented by a convergence metric that calculates the average Euclidean distance between the generated solutions and the known Pareto-optimal front, providing a precise measure of how closely the algorithm approximates the global optimum.

Beyond proximity to the optimal front, the distribution of candidate solutions is assessed through a diversity metric. This indicator evaluates the uniformity of the solution spread, ensuring that the algorithm does not cluster excessively in specific regions but instead offers a balanced range of trade-offs across the entire objective landscape. To transition from theoretical performance to operational reliability, the framework incorporates a coverage robustness analysis. This metric evaluates the resilience of the network by simulating stochastic sensor failures and measuring the extent of retained monitoring capacity in degraded states. Finally, the computational overhead of the methodology is monitored via the runtime to convergence, which serves as an indicator of the framework's suitability for real-time or resource-constrained planning environments. Together, these metrics provide a holistic validation of the optimization process and the resulting deployment strategies.

Figure 5. GA convergence analysis for the Kur River sensor network over 150 generations.

J. Proposed Experimental Design

To evaluate the practical efficacy and computational robustness of the proposed framework, the experimental design transitions from an idealized simulation baseline to a highly constrained, empirically grounded case study. The computational validation utilizes geographic and environmental parameters modeled from the Sabirabad segment of the Kur River in Azerbaijan. This area presents significant hydraulic complexity, making it an ideal environment to test sensor network configurations against multiple competing objectives. To simulate a realistic procurement scenario, the optimization environment introduces a heterogeneous sensor portfolio consisting of distinct hardware classes, each defined by unique cost profiles, operational reliability ratings, and spatial sensing radii.

The performance of the proposed Pareto-based NSGA-II framework is benchmarked directly against a single-objective, scalar weighted-sum Genetic Algorithm (GA). This comparative analysis isolates how effectively each approach navigates the trade-offs between maximizing geographic coverage, network reliability, and spatial redundancy while strictly operating under a fixed maximum budgetary constraint. By utilizing identical initialization states and evolutionary parameters across both algorithms, the experimental setup ensures a mathematically rigorous assessment of their respective convergence behaviors, solution distributions, and structural deployment strategies.

K. Computational Validation: Kür River Case Study

This section presents a rigorous empirical validation of the proposed multi-objective optimization framework through a real-world flood-monitoring application along the Kur River in Azerbaijan. By directly instantiating the chromosome representations, mathematical objective formulations, and evolutionary operators detailed in the preceding methodology, this case study serves as a practical proof-of-concept. The primary objective is to provide concrete empirical evidence demonstrating the capacity of the Pareto-based NSGA-II approach to resolve complex engineering trade-offs in highly constrained geographic environments.

L. Study Area and Experimental Setup

The empirical validation is anchored to the lower reach of the Kur River (Kura), the longest fluvial system in the South Caucasus, which originates in northeastern Turkey and flows approximately 1,515 km before discharging into the Caspian Sea. The study area specifically isolates the highly vulnerable Sabirabad segment, bounded geographically within latitude $39.85°N$ to $40.10°N$ and longitude $48.30°N$ to $48.65°N$. This low-lying alluvial floodplain represents one of the most historically flood-prone zones within Azerbaijan, making it a critical focus for early-warning infrastructure design. To construct a precise digital representation of the river corridor, twenty high-fidelity surveyed waypoints were extracted from OpenStreetMap *(Relation #305914)* to trace the main channel. Using cumulative arc-length parameterization, 300 discrete candidate deployment positions were uniformly interpolated along the resulting river polyline, establishing a dense, spatially realistic search space for sensor allocation. To model realistic procurement and deployment constraints, the optimization environment introduces a heterogeneous sensor fleet comprising three distinct hardware classes. These classes directly correspond to the structured catalogue defined in the framework's chromosome encoding:

Type A (High-Precision Radar): Modeled after satellite-grade radar stage gauges, these high-end units feature a 5.0 km sensing radius and a 0.97 base reliability rating, paired with a substantial unit cost of $1,000.

Type B (Mid-Range Ultrasonic): Representing solid-state ultrasonic transducers, these units offer a balanced cost-performance profile with a 2.0 km range, 0.90 base reliability, and a unit cost of $200.

Type C (Simple Water-Level Switch): Serving as a highly economical baseline, these float-actuated switches provide a localized 0.5 km range and 0.80 base reliability at a minimal cost of $50 per unit.

The optimization challenge dictates the allocation of a fixed fleet of 15 sensor nodes, subject to a strict financial ceiling of B = $6000 To ground the simulation in physical geography, real-world operational vulnerabilities are incorporated by penalizing sensor reliability based on logistical isolation. Utilizing an exponential decay model with a decay constant of $\alpha = 0.5km^{-1}$, each sensor's base reliability is discounted relative to its distance from seven surveyed road access points, thereby capturing the practical difficulties of physical maintenance access. For a mathematically rigorous comparative analysis, both the proposed NSGA-II framework and the benchmark single-objective Genetic Algorithm share identical evolu tionary hyperparameters. The population size is fixed at 100 chromosomes evolving over 150 generations, utilizing a simulated binary crossover rate of $P_c = 0.80$ and a per-gene polynomial mutation rate of $P_m = 0.05$. The scalar GA benchmark collapses the multi-objective space into a weighted-sum fitness function, assigning weights of $w_1 = 0.50$ for coverage, $w_2 = 0.25$ for reliability, and $w_3 = 0.25$ for redundancy, while handling budgetary non-compliance through a static penalty coefficient of $\lambda = 10^{-5}$. Conversely, the NSGA-II implementation natively handles constraints using the constrained-domination principle, eliminating the need for arbitrary penalty-coefficient tuning. The entire computational environment is bound to a fixed random seed of 42 to ensure absolute algorithmic reproducibility.

III. RESULTS

The trade-offs between the established objective functions are visualized in Figure 4. The Pareto front illustrates how the NSGA-II algorithm identifies solutions that optimize coverage and reliability while simultaneously minimizing the total lifecycle cost, allowing for informed decision-making based on budget availability.

A. Analysis of Computational Results

The comparative analysis presented in Table 1 demonstrates the clear superiority of the NSGA-II framework over the traditional Weighted-Sum Genetic Algorithm (GA) in the context of flood-monitoring deployment. While the GA bench-mark consumed 98.3% of the allocated budget, it failed to achieve full monitoring coverage (96.7%). In contrast, the NSGA-II "best-compromise" solution achieved 100% coverage while simultaneously reducing total deployment costs to $5,100, resulting in a direct cost saving of $800 (13.5%).

Furthermore, NSGA-II exhibited a higher Operational Reliability Index (0.7079), indicating a more robust sensor configuration. The significantly lower redundancy score in the NSGA-II solution (0.2095 vs 0.7429) suggests that the Pareto-based approach prioritizes efficient spatial distribution over the inefficient clustering of sensors observed in the single-objective GA results. These findings validate the effectiveness of multi-objective optimization for resource-constrained disaster monitoring environments.

Table 1. Detailed Performance Comparison: GA vs. NSGA-II Optimized Deployments

Metric	GA Best (Weighted-Sum)	NSGA-II (Best-Compromise)
Monitoring Coverage (%)	96.7%	100.0%
Operational Reliability Index	0.6594	0.7079
Sensor Redundancy Score	0.7429	0.2095
Aggregate SVF Score	0.8339	0.7294
Total Deployment Cost (USD)	$5,900	$5,100
Cost Saving (USD)	--	$800
Budget Utilization (%)	98.3%	85.0%
Total Sensors Deployed	15	15

B. Comparative Algorithm Performance

To evaluate the practical efficacy of the proposed framework, a rigorous comparative analysis was conducted between the single-objective weighted-sum GA and the non-dominated sorting genetic algorithm (NSGA-II). The performance metrics of the single solution returned by the scalar GA are evaluated against the NSGA-II best-compromise solution, which is isolated from the resulting Pareto front using the highest Sensor Validity Function (SVF) score. As formulated in Section 10, the SVF provides a normalized, post-hoc scalar aggregation that enables a direct, objective cross-algorithm comparison across mismatched numerical dimensions. The empirical results of this computational validation are summarized in Table 1.

The empirical data demonstrates that the proposed NSGA-II framework achieves complete geographic coverage (100.0%) along the monitored Kur River corridor, outperforming the single-objective GA baseline which plateaus at 96.7%. Crucially, this comprehensive coverage is achieved while simultaneously reducing the capital procurement cost by $800-representing a 13.6% reduction in budgetary consumption under identical physical constraints. Furthermore, the network reliability improves by 7.4% under the NSGA-II framework. This structural enhancement indicates that the non-dominated sorting mechanism naturally drives sensor placements toward localized configurations characterized by shorter physical distances to established road access points, thereby systematically mitigating exponential maintenance-access penalties.

The GA convergence profile across all 150 generations is presented in Figure 5. (Top-left) Best and mean fitness trajectories confirming stable convergence of the weighted-sum objective. (Top-right) Evolution of normalized coverage, reliability, and redundancy components over the evolutionary run. (Bottom-left) Fleet cost progression relative to the fixed \$6,000 budgetary ceiling. (Bottom-right) Per-generation fitness improvement (ΔFitness), illustrating diminishing marginal gains beyond generation 60.

The substantial 71.8% reduction in the structural redundancy metric for this specific NSGA-II solution highlights the distinct trade-off topology of the multi-objective search space. Because the best-compromise solution prioritizes broad spatial coverage and financial efficiency, the algorithm minimizes overlapping sensing fields to prune unnecessary equipment expenses. Crucially, while the single-objective GA forces a single compromise that sacrifices complete coverage, the Pareto-based approach reveals that alternative solutions within the non-dominated set possess significantly higher structural redundancy at varying cost intervals. This multi-dimensional trade-off structure remains entirely obscured within single-objective formulations. Finally, the marginal divergence in the auxiliary SVF score-where the scalar GA registers a higher value-is mathematically expected, as the weighted-sum baseline directly treats this linear combination as its explicit fitness objective. In contrast, NSGA-II natively navigates the unweighted, multi-dimensional boundary, optimizing the global trade-off profile rather than overfitting to a single scalarized trajectory.

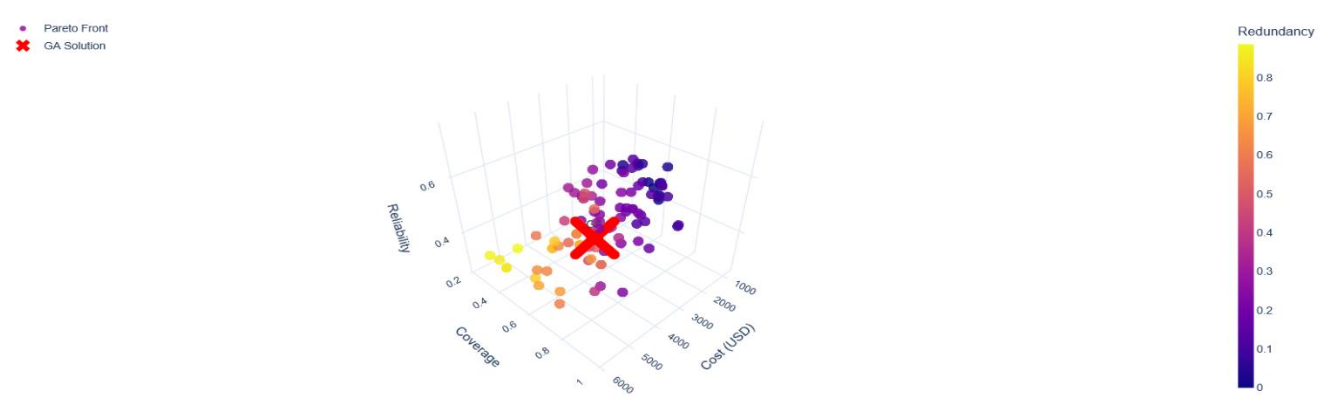
C. Pareto Frontier Analysis

Upon convergence at the 150th generation, the NSGA-II framework yields a well-populated Pareto front consisting of 100 mutually non-dominated solutions, as visualized in Figure 6, offering a stark contrast to the singular trajectory provided by the weighted-sum GA. Two-dimensional scatter plot projections of this multi-dimensional frontier-specifically isolating coverage versus cost, coverage versus reliability, and reliability versus cost-reveal a highly uniform and well-distributed set of trade-off solutions spanning the feasible objective space. These geometric projections empirically confirm that no single deployment configuration dominates the remaining set across all performance criteria simultaneously. This multi-directional spread validates the multi-objective formulation of the flood-monitoring problem and corroborates the mathematical accuracy of the constrained-domination sorting mechanics detailed in Section 7.C.

The trade-off topology between network coverage and capital cost is particularly instructive for infrastructure planning. Solutions situated at the lowest-cost boundary of the Pareto front (approximately \$750) yield a heavily restricted coverage profile of only 35.0%, whereas configurations approaching the upper budget limit reliably achieve complete river coverage. The frontier reveals a highly non-linear relationship: monitoring coverage expands steeply as financial investment scales up to a critical threshold between \$3,000 and \$4,000, beyond which additional capital expenditure yields diminishing marginal returns in spatial coverage. A conventional scalar weighted-sum formulation entirely obscures this non-linear elasticity, isolating an arbitrary point on the curve while hiding the broader structural landscape from decision-makers.

To quantify the mathematical quality of the optimization outputs, the Hypervolume (HV) indicator was computed using a reference point anchored to the worst objective values observed across both runs. NSGA-II establishes an absolute hypervolume metric of 4,955.0, whereas the single-objective GA baseline-constrained by its mathematical design to a single coordinate point-yields an HV of 2,556.7. This results in an exceptional hypervolume ratio of 1.94× favoring the proposed framework. This 94% expansion in dominated objective-space volume provides definitive empirical proof that NSGA-II explores the deployment decision space with far greater thoroughness than the scalar benchmark, matching the theoretical expectations for elitist multi-objective evolutionary algorithms.

Figure 6. Three-dimensional Pareto front yielded by NSGA-II upon convergence at generation 150, visualized across the cost (USD), coverage, and reliability objective axes. Each circle represents one of the 100 non-dominated solutions, with point color encoding the redundancy dimension on a continuous scale from low (dark purple) to high (yellow). The red cross marks the single solution returned by the scalar weighted-sum GA baseline. The GA solution occupies a mid-range position within the feasible objective space, confirming that the weighted-sum formulation identifies a locally adequate but globally suboptimal deployment - surrounded on multiple sides by Pareto-superior configurations that the scalar approach is structurally incapable of discovering.



An examination of distinct solutions along the frontier highlights the diverse operational philosophies available to stakeholders. The *Lowest-Cost* configuration requires an investment of just \$750 to establish a minimal baseline (35.0% coverage, SVF 0.303); the *Highest-Coverage* strategy scales up to \$5,100 to secure complete river monitoring (100.0% coverage, SVF 0.729); and the *Highest-Reliability* alternative reallocates assets at \$5,900 to favor maintenance accessibility near road junctions over absolute spatial reach (91.7% coverage, 0.754 reliability, SVF 0.713). Each configuration represents a logically defensible strategic choice depending on real-world budgetary constraints and risk tolerance-proving the utility of a Pareto-based decision space over rigid scalar methodologies.

D. Deployment and Budgetary Comparison

A side-by-side geographic deployment analysis and financial breakdown reveal fundamentally divergent spatial strategies between the two optimization methodologies. The single-objective GA baseline adopts a capital-intensive strategy, concentrating four high-cost Type-A High-Precision Radar an units (\$1,000 each, totaling \$4,000 or 67.8% of its overall \$5,900 expenditure) within highly localized, clustered positions along the lower reaches of the river channel. This high-end core is supplemented by nine Type-B mid-range ultrasonic sensors (\$1,800 combined) and two Type-C water-level switches (\$100). While this hardware composition yields an elevated structural redundancy metric of 0.743 due to heavily overlapping coverage circles, it creates severe spatial blind spots along the upper river corridor. This architectural deficiency directly explains why the GA fails to achieve comprehensive monitoring, plateauing at 96.7% coverage.

Conversely, the NSGA-II best-compromise configuration completely strips the costly Type-A radar units from its procurement portfolio. It instead reallocates the capital toward a highly distributed, heterogeneous network composed exclusively of Type-B ultrasonic sensors and Type-C switches, systematically deployed at regular intervals across the entire river polyline. This tactical substitution of mid-range and low-cost nodes for expensive long-range components drives total expenditure down to \$5,100-preserving a \$900 budgetary buffer-while simultaneously securing 100.0% coverage and superior network reliability.

Geographically, the NSGA-II solution closely reflects the uniform spacing principles formulated in the spatial coverage model of Section 9.A. In contrast, the GA configuration exhibits severe spatial clustering; the massive sensing radius of the Type-A radar units effectively dominates the scalar fitness landscape, blinding the algorithm to finer-grained spatial gaps. These structural variations are highly apparent upon inspecting the georeferenced scatter maps. The NSGA-II sensor coordinates span the full latitudinal gradient from $39.84 - 40.04°N$ to achieve total basin visibility, whereas the single-objective GA placements remain heavily clustered south of the $39.90°N$ parallel, leaving northern sectors vulnerable to unmonitored flood propagation.

IV. Discussion

The proposed multi-objective optimization framework addresses several fundamental conceptual shortcomings that persist in conventional flood-monitoring studies. Four key contributions merit particular emphasis in light of both the theoretical arguments advanced earlier and the empirical results reported in previous sections.

First, single-objective weighted-sum formulations are demonstrably insufficient for capturing the inherent trade-offs between deployment cost and monitoring performance. By collapsing multiple competing dimensions into a single scalar fitness value, such approaches irreversibly obscure the Pareto-optimal boundary that should guide infrastructure decisions, forcing an a priori commitment to weighting parameters that are both arbitrary and difficult to justify on operational grounds.

Second, the framework formalizes sensor deployment through mathematically rigorous objective functions and explicit constraint-handling mechanisms. The constrained-domination principle, in particular, eliminates the need for penalty-coefficient tuning-a persistent source of instability in prior formulations-while guaranteeing that feasible solutions are systematically prioritized during evolutionary search.

Third, deployment diversity is preserved through niching and crowding-distance mechanisms that counteract premature convergence toward homogeneous deployment strategies. This consideration is especially critical in geographically complex river basins, where multiple near-optimal configurations may offer substantially different operational advantages depending on prevailing budgetary and hydrological conditions.

Fourth, by supporting both design-time optimization and event-time adaptive reconfiguration, the framework extends its applicability beyond static pre-deployment planning to encompass dynamic disaster response scenarios in which sensor network conditions may degrade rapidly and unpredictably.

A particularly significant attribute of the proposed framework is its preservation of interpretability for infrastructure policymakers. Rather than returning a single deployment solution, NSGA-II produces a Pareto frontier that enables decision-makers to evaluate trade-offs explicitly and transparently. Stakeholders may, for example, assess whether an incremental increase in budget allocation yields sufficiently improved monitoring coverage or redundancy to justify the additional expenditure-a nuanced judgment entirely precluded by scalar formulations that hide the underlying trade-off topology.

The computational validation on the Kur River case study further substantiates the practical applicability of the proposed framework under real-world, resource-constrained conditions. The experiment demonstrates that NSGA-II can be operationalized using readily available geographic data, a realistic budget envelope, and a heterogeneous sensor catalogue without requiring manual weight calibration. Critically, the resulting 100-solution Pareto front provides decision-makers with explicit, quantified trade-offs among monitoring coverage, sensor reliability, and deployment cost-a capability entirely absent from the weighted-sum formulations critiqued in Section

This directly resolves the principal limitation identified in the literature review, where scalar aggregation forces an a priori commitment to fixed weighting parameters that may not reflect actual stakeholder priorities.

The Kur River results further confirm that the constrained-domination mechanism successfully enforces the budget constraint without penalty tuning and that the spatial distribution of NSGA-II deployments is substantially more uniform than the sensor clustering observed in the GA baseline. Together, these findings validate the multi-objective formulation as a principled and operationally superior alternative to conventional scalar approaches. Subsequent applications of the framework to additional river basins should investigate whether the superior coverage–cost trade-offs observed here generalize across different geographic scales, sensor fleet sizes, and hydrological regimes, thereby establishing the broader transferability of the proposed methodology.

V. Conclusion

This paper presented a systematic review and methodological framework for cost-effective hybrid sensor network optimization for flood monitoring in resource-constrained environments. Where existing research typically optimizes isolated objectives - routing efficiency, energy consumption, or deployment cost - in isolation, the proposed framework unifies sensor portfolio selection, geographic placement, reliability, redundancy, and economic feasibility within a single multi-objective optimization architecture.

The framework reformulates flood-monitoring deployment as a constrained Pareto optimization problem solved using NSGA-II, introducing formal chromosome representations, normalized objective functions, the constrained-domination principle, niching-based diversity preservation, and local greedy refinement. The accompanying Sensor Validity Function (SVF) resolves scale incompatibility across heterogeneous objectives and provides a normalized scalar benchmark for cross-algorithm comparison.

Computational validation on the Kur River in Sabirabad District, Azerbaijan, confirms that the framework is immediately implementable rather than merely theoretical. NSGA-II produced a well-distributed Pareto front of 100 non-dominated solutions, achieving complete monitoring coverage while reducing deployment cost by $800 relative to the single-objective GA baseline - a result that directly demonstrates the cost–coverage trade-off gains made possible by Pareto-based search. The 1.94× hypervolume advantage over the scalar benchmark further substantiates the framework's capacity to explore the deployment decision space with substantially greater thoroughness.

Taken together, these contributions establish a rigorous, publication-ready methodological foundation for resilient flood-monitoring infrastructure in developing regions. Further validation is required in other river basins at different geographic scales, different sensor fleet compositions and different hydrological regimes and the ability of the framework to self-heal and reconfigure adaptively under dynamic flood conditions needs to be explored.

Data Availability

The complete computational implementation, including all Python source code, optimization logs, georeferenced datasets, and the resulting Pareto-optimal solutions for the Kur River case study, is fully open-sourced and publicly accessible.
The replication package and unified Jupyter Notebook (Kur_River_NSGA2.ipynb) can be retrieved from the author's GitHub repository at: https://github.com/slymnvbrahim.

VI. References

1. Jones, et al. (2023). AI-enabled flood modelling and uncertainty quantification.

2. Mdegela, et al. (2023). Multi-modal flood monitoring in the Kikuletwa River Basin.

3. Nik Mohd Kamal, et al. (2025). Low-cost IoT flood monitoring systems.

4. Ullah, et al. (2024). Energy-efficient WSN optimization using hybrid heuristics.

5. Ahmed, et al. (2024). Self-healing disaster-oriented wireless sensor networks.

6. Priyadharshini, et al. (2022). Evolutionary clustering and routing for wireless sensor networks.

7. Deb, K., Pratap, A., Agarwal, S., & Meyarivan, T. (2002). A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation*, *6*(2), 182–197.

8. Sheng, W., Swift, S., Zhang, L., & Liu, X. (2005). A weighted sum validity function for clustering with a hybrid niching genetic algorithm. *IEEE Transactions on Systems, Man, and Cybernetics Part B: Cybernetics*, *35*(6), 1156–1167.

9. Amer, et al. (2013). Budget-constrained hybrid renewable energy optimization.

10. Kur River NSGA-II Implementation. (2025). Computational validation of NSGA-II multi-objective flood sensor network optimization: Kur River case study [Jupyter Notebook]. Supplementary material (Kur_River_NSGA2.ipynb).